Week 05 teacher guide

Decimals and long division

PowerPoint slides 395-485 | Student workbook pp. 38-45

Decimal digits name parts of a whole. Addition and subtraction combine equal place-value units. Division preserves its quotient when both quantities scale equally.

The seven days

Day 029 · p. 38 · Saturday · 20 questions,including 2 challenges
Decimal place value and products

Day 030 · pp. 39-40 · Sunday · 20 questions,including 2 challenges
Long division with decimals

Day 031 · p. 41 · Monday · 20 questions,including 2 challenges
Decimals and long division

Day 032 · p. 42 · Tuesday · 20 questions,including 2 challenges
Decimals and long division

Day 033 · p. 43 · Wednesday · 25 questions,including 2 challenges
Decimals and long division

Day 034 · p. 44 · Thursday · 20 questions,including 2 challenges
Decimals and long division

Day 035 · p. 45 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

The student pages contain questions only. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons use 20 core questions including C1 and C2. Weekdays use 20 questions; Wednesday uses 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The Sunday reference page is for checking a method after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer-review sections are optional after Friday and do not add to the 80-minute weekend lesson.

Built-in review

Day 031: Decimal practice (9, 10, 11, 12)

Day 032: Multiplication and long division (9, 10, 11, 12)

Day 033: Weekly puzzle (13)

Day 034: Fraction operations (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Saturday. Decimal place value and products

Day 029 · p. 38 | Slides 395-424

Goal: I can add, subtract, and multiply decimals while keeping track of the value of each digit.

Check first: Whole-number multiplication, regrouping, and decimal place names.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 395-396 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 397-398 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 399-400 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 401 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 402-403 · Explore · P4, P5, P6

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 404-406 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 407 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 408 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 409 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 410 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 411-414 · Guided practice · X1, X2, X3, X4

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 415-416 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 417-420 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 421 · Summary · S

Ask for one sentence or annotated example in the learner's own words. Listen for the main relationship.

Slides 422 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 423 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 424 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: Place value explains the product

Watch and discuss; this is a projected investigation, not an extra assigned question.

Compare 0.3×0.7 with 0.3×0.07. Predict the factor-of-ten relationship, then explain why the products use hundredths and thousandths. Contrast both with $0.3 + 0.07$.

1. Predict how changing 0.7 to 0.07 changes the product.
2. Multiply counts $3 \times 7 = 21$ in both examples.
3. Name the product units: hundredths or thousandths.
4. Contrast with addition by renaming both addends in hundredths.

Ask: Why do addition and multiplication treat the decimal places differently?

If stuck: Name the size of one unit in each factor before counting how many units are in the product.

Expected: 0.21 and 0.021; the second product is one tenth of the first. The sum is 0.37 because addition combines equal place values rather than multiplying unit sizes.

C1 · Challenge discussion

Ask: Which estimate would catch a misplaced decimal without giving the exact answer?

If stuck: Think of 2.4 meters as slightly less than $2\frac{1}{2}$ meters at slightly less than \$4 per meter.

Expected: Cost \$9.00; change \$11.00

C2 · Challenge discussion

Ask: Can you predict the smaller area before doing the second multiplication?

If stuck: Notice that A's area becomes one factor in B's area calculation.

Expected: A: 0.14 m²; B: 0.126 m²; difference 0.014 m²

What to notice

Aligns decimal points when adding. → Ask the student to name each column.

Estimates product size before computing. → Ask whether taking a fraction of a fraction can exceed either factor.

Common error: Aligns the rightmost digits. Repair: Write zeros to make place-value columns visible.

Sunday. Long division with decimals

Day 030 · pp. 39-40 | Slides 425-451

Goal: I can divide with decimals by preserving equal groups and checking my quotient.

Check first: Whole-number multiplication and decimal place value.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 425-426 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 427-428 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 429-430 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 431 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 432-433 · Explore · P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 434-436 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 437 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 438 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 439 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 440 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 441 · Explore · M1, M2, M3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 442-443 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 444-447 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 448 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 449 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 450 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 451 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Sunday. Discussion and support

Investigation T: Which rescaling keeps the quotient?

Watch and discuss; this is a projected investigation, not an extra assigned question.

Start with 0.48 ÷ 0.06. Compare 4.8 ÷ 0.6, 0.48 ÷ 0.6, and 4.8 ÷ 0.06. Predict each change before calculating; explain using the sizes and numbers of groups.

1. Interpret the original quotient as a count of 0.06-sized groups.
2. Scale both quantities by 10 and explain why the count stays 8.
3. Scale only the divisor and predict 0.8.
4. Scale only the dividend and predict 80; check all results by multiplication.

Ask: Is moving the decimal point a reason, or a shortcut for changing units? Which quantities must be rescaled together?

If stuck: Draw the original total and group size, then identify which one changes in each row.

Expected: 8, 8, 0.8, and 80, respectively. Scaling both amounts preserves the quotient; scaling only the divisor divides it by 10; scaling only the dividend multiplies it by 10.

C1 · Challenge discussion

Ask: Why do the two divisions use different units?

If stuck: First count bottles from liters; then count boxes from bottles.

Expected: 32 filled bottles; 5 full boxes; 2 bottles remain

C2 · Challenge discussion

Ask: How can you check the remainder both by length and by strip count?

If stuck: Find the complete strip count before subtracting the strips used.

Expected: 21 strips originally; 0.70 meter remains; 2 unused strips

What to notice

Scales both numbers equally. → Ask for the factor written beside each number.

Adds zeros after a decimal point to continue. → Ask what three remaining ones become in tenths.

Common error: Moves only one decimal point. Repair: Rename both quantities in the same unit.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

395. Week 5, Saturday
Watch and discuss

396. Today
Watch and discuss

397. Recall
Day 029 · Workbook p. 38 · R1, R2, R3, R4

398. Recall: answers
Day 029 · Workbook p. 38 · R1, R2, R3, R4

399. Different decimal lengths
Day 029 · Workbook p. 38 · P1

400. Subtract across zeros
Day 029 · Workbook p. 38 · P2, P3

401. The model
Watch and discuss

402. Product below both factors
Day 029 · Workbook p. 38 · P4

403. Leading zeros in a product
Day 029 · Workbook p. 38 · P5, P6

404. Worked example · 1 of 3
Watch and discuss · worked example

405. Worked example · 2 of 3
Watch and discuss · worked example

406. Worked example · 3 of 3
Watch and discuss · worked example

407. Your turn
Day 029 · Workbook p. 38 · F

408. Two ways to see it
Watch and discuss · compare the same example

409. A claim to check
Day 029 · Workbook p. 38 · D

410. Words and meaning
Watch and discuss

411. Guided practice · X1
Day 029 · Workbook p. 38 · X1

412. Guided practice · X2
Day 029 · Workbook p. 38 · X2

413. Guided practice · X3
Day 029 · Workbook p. 38 · X3

414. Guided practice · X4
Day 029 · Workbook p. 38 · X4

415. Place value explains the product
Watch and discuss · draw on separate paper

416. Place value explains the product
Watch and discuss · draw on separate paper

417. Challenge C1
Day 029 · Workbook p. 38 · C1

418. Challenge C1 · Discuss
Day 029 · Workbook p. 38 · C1

419. Challenge C2
Day 029 · Workbook p. 38 · C2

420. Challenge C2 · Discuss
Day 029 · Workbook p. 38 · C2

421. In your own words
Day 029 · Workbook p. 38 · S

422. Exit ticket
Day 029 · Workbook p. 38 · E

423. The week ahead
Days 031-035 · Workbook pp. 41-45

424. Tutor reference
Tutor only;hidden in slide show

Sunday

425. Week 5, Sunday
Watch and discuss

426. Today
Watch and discuss

427. Recall
Day 030 · Workbook p. 39 · R1, R2, R3, R4

428. Recall: answers
Day 030 · Workbook p. 39 · R1, R2, R3, R4

429. Three-digit by two-digit exact quotient
Day 030 · Workbook p. 39 · P1

430. Whole numbers with decimal quotient
Day 030 · Workbook p. 39 · P2, P3

431. The model
Watch and discuss

432. Decimal divided by decimal
Day 030 · Workbook p. 39 · P4, P5

433. Divisor below one makes larger quotient
Day 030 · Workbook p. 39 · P6, P7

434. Worked example · 1 of 3
Watch and discuss · worked example

435. Worked example · 2 of 3
Watch and discuss · worked example

436. Worked example · 3 of 3
Watch and discuss · worked example

437. Your turn
Day 030 · Workbook p. 39 · F

438. Two ways to see it
Watch and discuss · compare the same example

439. A claim to check
Day 030 · Workbook p. 39 · D

440. Words and meaning
Watch and discuss

441. Mixed review
Day 030 · Workbook p. 39 · M1, M2, M3

442. Which rescaling keeps the quotient?
Watch and discuss · draw on separate paper

443. Which rescaling keeps the quotient?
Watch and discuss · draw on separate paper

444. Challenge C1
Day 030 · Workbook p. 39 · C1

445. Challenge C1 · Discuss
Day 030 · Workbook p. 39 · C1

446. Challenge C2
Day 030 · Workbook p. 39 · C2

447. Challenge C2 · Discuss
Day 030 · Workbook p. 39 · C2

448. In your own words
Day 030 · Workbook p. 39 · S

449. Exit ticket
Day 030 · Workbook p. 39 · E

450. The week ahead
Days 031-035 · Workbook pp. 41-45

451. Tutor reference
Tutor only;hidden in slide show

Answer review

452. Week 5 answers
Next lesson: slide 486. Use this review section after Friday.

453. Decimal place value and products · 1 of 5
Day 029 · Workbook p. 38 · Items R1, R2, R3, R4, P1

454. Decimal place value and products · 2 of 5
Day 029 · Workbook p. 38 · Items P2, P3, P4, P5, P6

455. Decimal place value and products · 3 of 5
Day 029 · Workbook p. 38 · Items F, D, X1, X2, X3

456. Decimal place value and products · 4 of 5
Day 029 · Workbook p. 38 · Items X4, S, E, C1

457. Decimal place value and products · 5 of 5
Day 029 · Workbook p. 38 · Items C2

458. Long division with decimals · 1 of 5
Day 030 · Workbook p. 39 · Items R1, R2, R3, R4, P1

459. Long division with decimals · 2 of 5
Day 030 · Workbook p. 39 · Items P2, P3, P4, P5, P6

460. Long division with decimals · 3 of 5
Day 030 · Workbook p. 39 · Items P7, F, D, M1, M2

461. Long division with decimals · 4 of 5
Day 030 · Workbook p. 39 · Items M3, S, E, C1

462. Long division with decimals · 5 of 5
Day 030 · Workbook p. 39 · Items C2

463. Decimals and long division · 1 of 4
Day 031 · Workbook p. 41 · Items 1, 2, 3, 4, 5

464. Decimals and long division · 2 of 4
Day 031 · Workbook p. 41 · Items 6, 7, 8, 9, 10

465. Decimals and long division · 3 of 4
Day 031 · Workbook p. 41 · Items 11, 12, 13, 14, 15

466. Decimals and long division · 4 of 4
Day 031 · Workbook p. 41 · Items 16, 17, 18, C1, C2

467. Decimals and long division · 1 of 4
Day 032 · Workbook p. 42 · Items 1, 2, 3, 4, 5

468. Decimals and long division · 2 of 4
Day 032 · Workbook p. 42 · Items 6, 7, 8, 9, 10

469. Decimals and long division · 3 of 4
Day 032 · Workbook p. 42 · Items 11, 12, 13, 14, 15

470. Decimals and long division · 4 of 4
Day 032 · Workbook p. 42 · Items 16, 17, 18, C1, C2

471. Decimals and long division · 1 of 5
Day 033 · Workbook p. 43 · Items 1, 2, 3, 4, 5

472. Decimals and long division · 2 of 5
Day 033 · Workbook p. 43 · Items 6, 7, 8, 9, 10

473. Decimals and long division · 3 of 5
Day 033 · Workbook p. 43 · Items 11, 12, 13, 14, 15

474. Decimals and long division · 4 of 5
Day 033 · Workbook p. 43 · Items 16, 17, 18, 19, 20

475. Decimals and long division · 5 of 5
Day 033 · Workbook p. 43 · Items 21, 22, 23, C1, C2

476. Decimals and long division · 1 of 5
Day 034 · Workbook p. 44 · Items 1, 2, 3, 4, 5

477. Decimals and long division · 2 of 5
Day 034 · Workbook p. 44 · Items 6, 7, 8, 9

478. Decimals and long division · 3 of 5
Day 034 · Workbook p. 44 · Items 10, 11, 12, 13

479. Decimals and long division · 4 of 5
Day 034 · Workbook p. 44 · Items 14, 15, 16, 17, 18

480. Decimals and long division · 5 of 5
Day 034 · Workbook p. 44 · Items C1, C2

481. Weekly check · 1 of 5
Day 035 · Workbook p. 45 · Items 1, 2, 3, 4, 5

482. Weekly check · 2 of 5
Day 035 · Workbook p. 45 · Items 6, 7, 8, 9

483. Weekly check · 3 of 5
Day 035 · Workbook p. 45 · Items 10, 11, 12, 13, 14

484. Weekly check · 4 of 5
Day 035 · Workbook p. 45 · Items 15, 16, 17, 18, C1

485. Weekly check · 5 of 5
Day 035 · Workbook p. 45 · Items C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 029 · Decimal place value and products

Workbook p. 38 · assigned set

R1 · R-TABLE

16

R2 · P-WHOLE

60 shells

R3 · F-DIV

$\frac{8}{9}$ liter

R4 · R-TABLE

21

P1 · D-ADD

12.736

P2 · D-SUB

7.315

P3 · D-MULT

$9.40

P4 · D-MULT

0.24; yes

P5 · D-MULT

0.003

P6 · D-MONEY

$4.48

F · D-MULT

thousandths; 0.021

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · D-MULT

Six tenths times four tenths is twenty-four hundredths, or 0.24.

Reason: He counts digits without tracking the size of the units. Shade six tenths across and four tenths down on a hundred-square grid.

X1 · D-ADD

2.854

X2 · D-SUB

11.819

Reason: Write 12.000. Regroup through the zero columns and subtract equal place values.

X3 · D-MULT

0.03

Reason: 6 tenths times 5 hundredths produces thousandths. The answer must be below either factor because both are below 1.

X4 · D-MULT

$12.95

S · EXPLAIN

Answers vary. Addition combines like-sized units. Multiplication combines both the number of parts and the size of each part.

Reason: Accept an accurate explanation supported by a matching example.

E · D-MULT

0.027

Reason: 45 × 6 = 270. Hundredths times hundredths makes ten-thousandths: 0.0270. It is less than 0.06.

C1 · D-MONEY

Cost $9.00; change $11.00

Reason: Compute 3.75 × 2.4 = 9.00. Subtract from 20 to get 11. An estimate of $4 × 2.5 = $10 makes $9 reasonable; answers such as $0.90 or $90 would not fit.

C2 · D-MULT

A: 0.14 m²; B: 0.126 m²; difference 0.014 m²

Reason: A's area is 0.4 × 0.35 = 0.14. B's area is 0.14 × 0.9 = 0.126, which is $\frac{9}{10}$ of 0.14. Subtract to get 0.014 m².

Multiplying a positive amount by 0.9 reduces it.

Day 030 · Long division with decimals

Workbook p. 39 · assigned set

R1 · R-EQUIV

10 blue tiles

R2 · R-TABLE

0, 12, and 21 centimeters

R3 · F-DIV

3

R4 · R-TABLE

16

P1 · D-DIV

28

P2 · D-DIV

3.375

P3 · D-DIV

1.2

P4 · D-DIV

70

Reason: Multiply both amounts by 100: 630 ÷ 9 = 70.

P5 · D-DIV

0.35

P6 · D-DIV

9 pieces

P7 · N-DIV

42

Reason: 28 × 40 = 1,120. Two more groups make 56, so 42 groups make 1,176.

F · D-DIV

540; 180

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · D-DIV

Both amounts must scale by 100. Use 630 ÷ 9 = 70.

Reason: She changes the group size without changing the available amount by the same factor. Compare 63 tenths with nine hundredths by renaming both in hundredths.

M1 · R-WHOLE

$\frac{3}{8}$

M2 · P-PART

1.8000 dollars

M3 · F-DIV

$3\frac{1}{3}$

S · EXPLAIN

Answers vary. Multiplying the dividend and divisor by the same power of ten preserves the quotient.

Reason: Accept an accurate explanation supported by a matching example.

E · D-DIV

120

Reason: Multiply both by 100: 840 ÷ 7 = 120. Check 120 × 0.07 = 8.4.

C1 · D-DIV1

32 filled bottles; 5 full boxes; 2 bottles remain

Reason: Scale 4.8 ÷ 0.15 by 100 to get 480 ÷ 15 = 32 bottles. Divide 32 by 6: five boxes use 30 bottles, and 2 remain. All 32 bottles account for 4.8 liters.

C2 · D-DIV1

21 strips originally; 0.70 meter remains; 2 unused strips

Reason: Multiply both numbers in 7.35 ÷ 0.35 by 100 : 735 ÷ 35 = 21. Nineteen strips use 6.65 meters; 7.35 − 6.65 = 0.70 meter. That remainder holds two 0.35-meter strips.

Day 031 · Decimals and long division

Workbook p. 41 · assigned set

1 · N-FACT

21

2 · N-FACT

49

3 · N-FACT

28

4 · N-FACT

63

5 · F-DIV

$\frac{5}{24}$ liter

6 · R-TABLE

32 laps

7 · N-FACT

24

8 · D-ADD

4.528

9 · D-ADD · review

6.528

10 · D-MULT · review

0.0050

11 · D-SUB · review

7.714

12 · D-MULT · review

0.56; yes

13 · D-MULT

Six tenths times four tenths is twenty-four hundredths, or 0.24.

14 · D-ADD
7.292

15 · D-SUB
14.413

Reason: Write 15.000. Regroup through the zero columns and subtract equal place values.

16 · D-MULT
0.018

Reason: 6 tenths times 3 hundredths produces thousandths. The answer must be below either factor because both are below 1.

17 · D-MULT
$11.85

18 · D-MONEY
$7.38

C1 · D-MONEY
Enough; change $0.20

Reason: Three notebooks cost 3 × 2.65 = $7.95. Adding $1.85 gives $9.80. Ten dollars exceeds that total by $0.20. Keeping cents aligned avoids interpreting 0.20 as 20 dollars.

C2 · D-MULT
0.25 × 0.12 = 0.03; 0.25 × 1.2 = 0.3

Reason: Twenty-five hundredths times twelve hundredths is 300 ten-thousandths, or 0.03. Also 0.25 is one quarter, and a quarter of 0.12 is 0.03. Increasing the second factor tenfold increases the product tenfold to 0.3.

Day 032 · Decimals and long division

Workbook p. 42 · assigned set

1 · F-EQUIV
0.5; 50%

2 · F-EQUIV
0.25; 25%

3 · F-EQUIV
0.75; 75%

4 · F-EQUIV
0.125; 12.5%

5 · P-WHOLE
60 shells

6 · F-SUB
$\frac{5}{12}$

7 · N-FACT
12

8 · D-DIV
20

9 · N-DIV · review
13

10 · N-MULT · review
1404

11 · N-DIV · review
42

12 · N-DIV · review
13

13 · D-DIV
Both amounts must scale by 100. Use 630 ÷ 9 = 70.

14 · D-DIV1
76

Reason: Multiply both numbers by 100 : 304 ÷ 4. The quotient stays 76.

15 · D-DIV1
3.28

16 · N-DIV1
8

Reason: 12 × 8 = 96, so the quotient is 8.

17 · R-RATE
$0.40 per marker

18 · D-ADD
3.171

C1 · D-DIV1
40 pieces at 0.24 meter; 32 pieces at 0.3 meter

Reason: Scale both divisions to whole-number divisors: 960 ÷ 24 = 40 and 96 ÷ 3 = 32. A smaller piece size fits more times into the same positive total length.

C2 · D-DIV1
27 km/h; 32.4 km in 1.2 hours

Reason: Scale 18.9 ÷ 0.7 to 189 ÷ 7 = 27 km/h. Then multiply 27 × 1.2 = 32.4 km. Units change from distance per hour times hours to distance.

Day 033 · Decimals and long division

Workbook p. 43 · assigned set

1 · N-SCALE
72

2 · N-SCALE
7

3 · N-SCALE
15

4 · N-SCALE
21

5 · D-DIV
30

6 · D-MULT
0.0075

7 · P-PART
1.8000 dollars

8 · P-WHOLE
80 shells

9 · F-SUB
$\frac{5}{12}$

10 · R-EQUIV
4 : 7

11 · N-FACT
15

12 · G-VOL
45 cm³

13 · PUZZLE · review
4.82

Reason: A ones digit of 4 places the number nearest 5. Among 4.28 and 4.82, the larger is nearer to 5.

14 · D-ADD
4.122

15 · D-SUB
15.703

Reason: Write 16.000. Regroup through the zero columns and subtract equal place values.

16 · D-MULT
0.027

Reason: 3 tenths times 9 hundredths produces thousandths. The answer must be below either factor because both are below 1.

17 · D-MULT
$12.25

18 · D-MONEY
$7.18

19 · D-DIV1
40

Reason: Multiply both numbers by 100 : 240 ÷ 6. The quotient stays 40.

20 · D-DIV1
1.84

21 · N-DIV1
20

Reason: 24 × 20 = 480, so the quotient is 20.

22 · R-RATE
$0.60 per marker

23 · F-DIV1
$\frac{2}{3}$

C1 · D-DIV1
22 full bags; 0.30 kilogram remains

Reason: The available amount is 12 − 1.8 = 10.2 kilograms. 10.2 ÷ 0.45 = 22 $\frac{2}{3}$ bagfuls. Twenty-two bags use 22 × 0.45 = 9.9 kilograms, leaving 0.3 kilogram. That remainder is less than one 0.45-kilogram bag.

C2 · D-DIV1
First:$3.00/m and $12 for 4 m; second:$3.20/m and $12.80 for 4 m

Reason: Divide each price by length: 13.50 ÷ 4.5 = 3, and 10.24 ÷ 3.2 = 3.2. Multiply each unit price by 4. The first board has the lower price per meter even though its total price is higher.

Day 034 · Decimals and long division

Workbook p. 44 · assigned set

1 · N-EST
700

2 · N-EST
8

3 · N-EST
About 100

4 · N-EST
About 20

5 · D-DIV
30

6 · D-MULT
0.0075

7 · D-DIV

0.875

8 · F-DIV

2

9 · F-DIV · review

20; $\frac{1}{20}$

10 · F-MULT · review

$2\frac{2}{3}$

11 · F-MULT · review

$\frac{4}{15}$

12 · F-MULT · review

$\frac{4}{5}$

13 · D-MONEY

$2.57

14 · D-DIV

Both amounts must scale by 100. Use 630 ÷ 9 = 70.

15 · D-ADD

6.341

16 · D-SUB

11.5

Reason: Write 12.000. Regroup through the zero columns and subtract equal place values.

17 · D-MULT

0.024

Reason: 3 tenths times 8 hundredths produces thousandths. The answer must be below either factor because both are below 1.

18 · D-MULT

$21.00

C1 · D-MONEY

$61.40 remains;$13.60 more is needed

Reason: Revenue is 24 × 3.75 = $90. Subtract $28.60 to leave $61.40. Compare that result with $75: the shortfall is $13.60.

C2 · D-DIV1

9 square meters; 100 tiles

Reason: The floor area is 3.6 × 2.5 = 9 m². Divide the area by one tile's area: 9 ÷ 0.09 = 100 tiles. Check 100 × 0.09 = 9.

Day 035 · Weekly check

Workbook p. 45 · assigned set

1 · D-DIV

30

2 · D-MULT

0.0075

3 · D-DIV

0.875

4 · D-MULT

0.0100

5 · F-DIV

$\frac{5}{24}$ liter

6 · R-RATE

4 labels per minute; $\frac{1}{4}$ minute per label

7 · F-DIV

$\frac{3}{4}$

8 · F-SUB

$\frac{5}{12}$

9 · R-EQUIV

4 : 7

10 · D-MULT

Six tenths times four tenths is twenty-four hundredths, or 0.24.

11 · D-ADD

8.56

12 · D-SUB

18.297

Reason: Write 19.000. Regroup through the zero columns and subtract equal place values.

13 · D-MULT

0.021

Reason: 3 tenths times 7 hundredths produces thousandths. The answer must be below either factor because both are below 1.

14 · D-MULT

$31.85

15 · D-MONEY

$6.66

16 · D-DIV1

10

Reason: Multiply both numbers by 100 : 30 ÷ 3. The quotient stays 10.

17 · D-DIV1

3.2

18 · N-DIV1

10

Reason: 14 × 10 = 140, so the quotient is 10.

C1 · D-DIV1

The first two both equal 12; 0.84 ÷ 0.7 = 1.2, one tenth as large

Reason: Scaling both dividend and divisor by 10 preserves the quotient. Scaling only the divisor by 10 makes each group ten times larger, so only one tenth as many groups fit. Calculations are 84 ÷ 7 = 12 and 8.4 ÷ 7 = 1.2.

C2 · D-MULT

0.70 kilogram remains; no, 4 loaves need 0.72 kilogram, so 0.02 kilogram is missing

Reason: Thirty-five loaves use 35 × 0.18 = 6.3 kilograms. 7 − 6.3 = 0.7 kilogram remains. Four loaves need 4 × 0.18 = 0.72 kilogram; the remainder falls short by 0.02 kilogram.